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Plant Kingdom

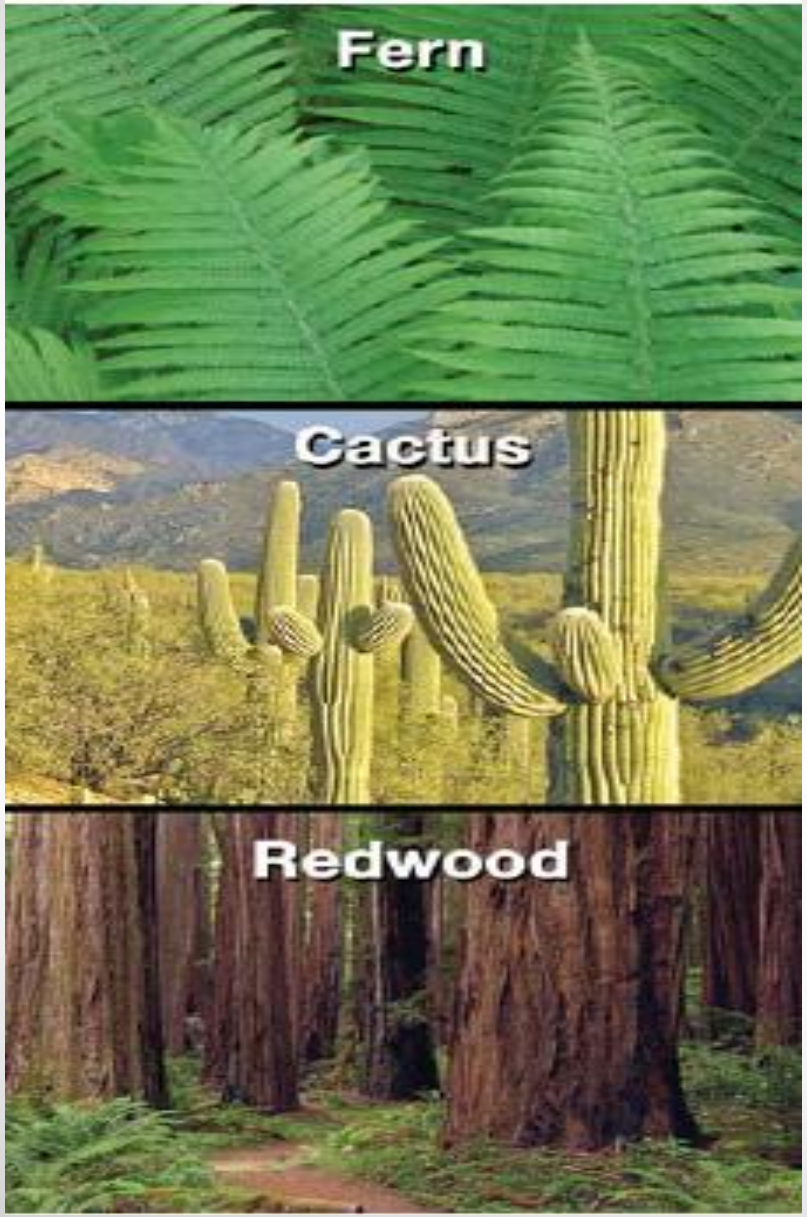


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Plant Characteristics

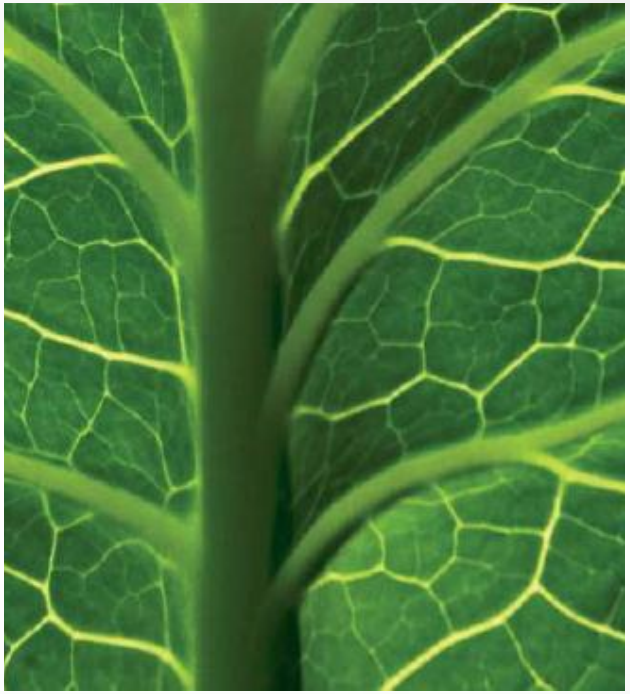


- Plants come in all sizes, from the tiny duckweed which grows to only about 10 mm in length, to the giant redwood which grows to about 100 m in height.

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Plant Characteristics

- All plants share the following characteristics:
  1. **Plants are producers and use photosynthesis to make food.**
  2. **Plants have eukaryotic cells with cell walls.**
  3. **Plants have a cuticle.**



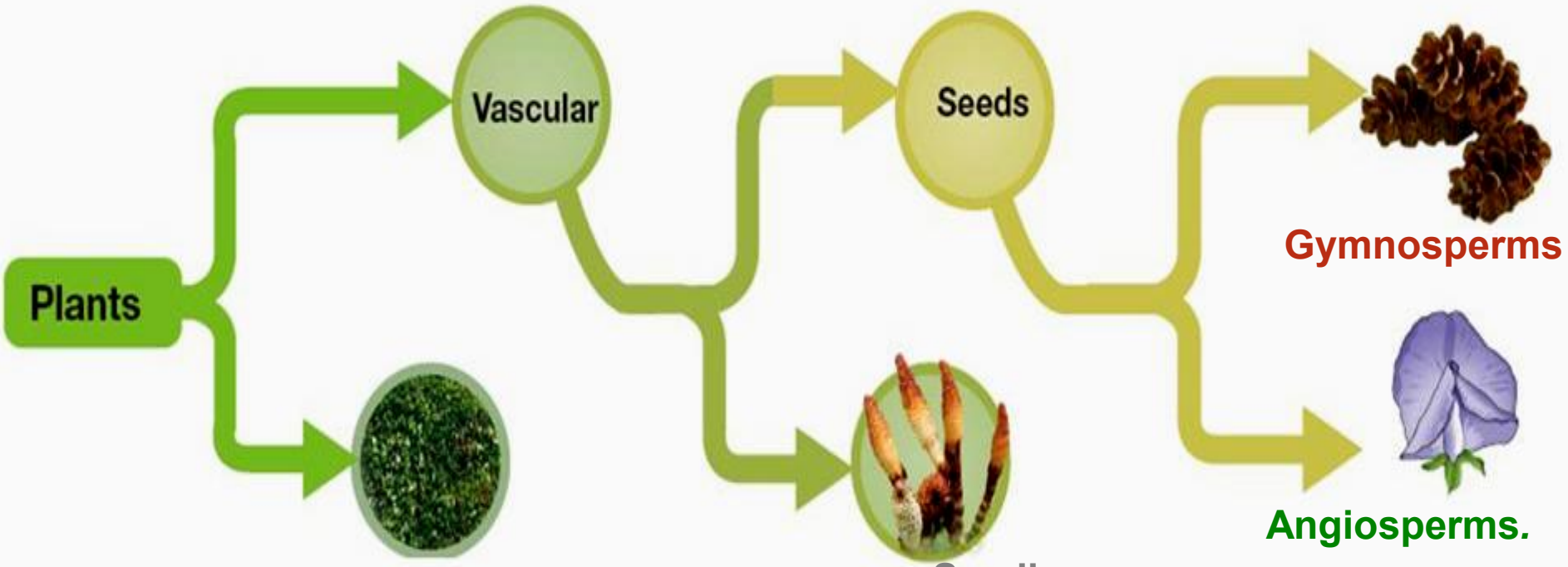
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Classification of Kingdom Plantae

- The plant kingdom is further classified into subgroups. Classification is based on the following criteria:
- **Plant body**: Presence or absence of a well-differentiated plant body. e.g., root, stem, and leaves.
- **Vascular system**: Presence or absence of a vascular system for the transportation of water and other substances. e.g., phloem and xylem.
- **Seed formation**: Presence or absence of flowers and seeds, and whether the seeds are naked or enclosed in a fruit.

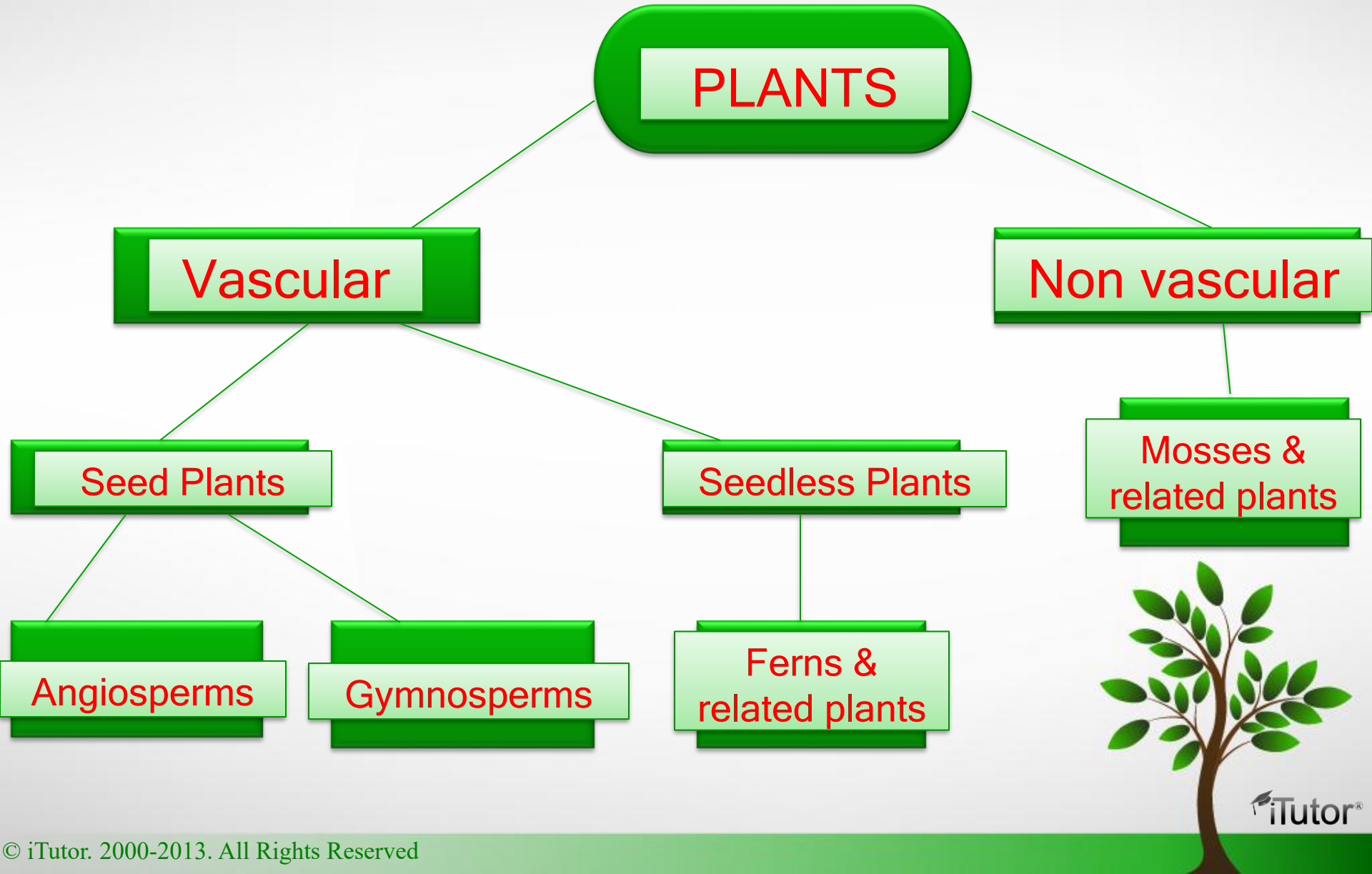


Plant Characteristics



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Plant Characteristics



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Plant Characteristics



1) Vascular Plants:
   - ❖ Broken into 2 more subgroups called:
     a) Seedless plants
     b) Seed plants
2) Nonvascular Plants
   - ❖ Not broken into subgroups

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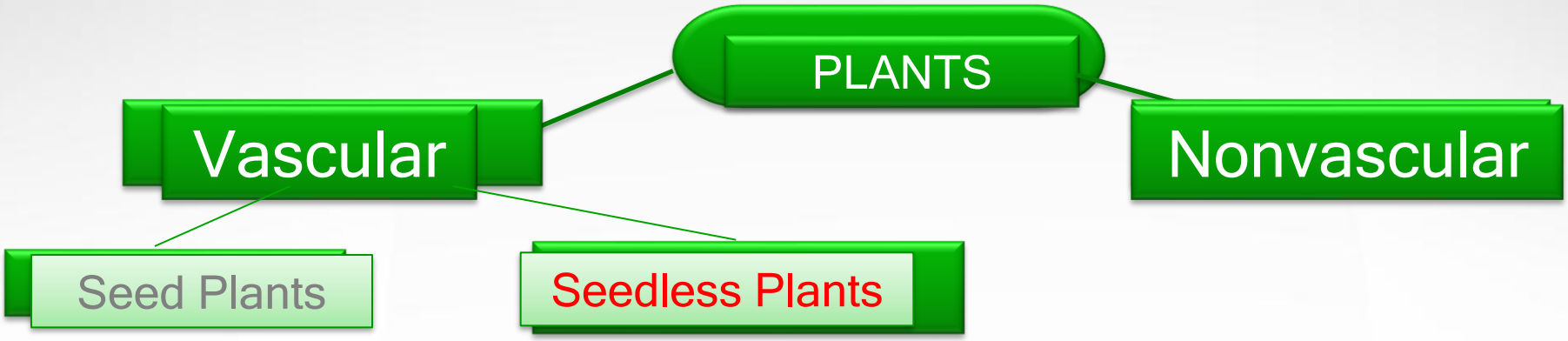
Vascular plants

- Vascular plants are divided into two groups–those that produce seeds and those that do not.
- Plants that produce seeds are divided into gymnosperms and angiosperms.
- Examples of plants that have no seeds are ferns, mosses and horsetails.



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Vascular plants



1) Seedless plants
   - Vascular plant that **DO NOT** have seeds
     - Ex: Ferns
2) Seed Plants
   - **Have** seeds for reproduction
     - Ex: conifers & flowering plants

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Seedless Vascular plants

- The seedless vascular plants include ferns, club mosses, and horsetails.



Ferns



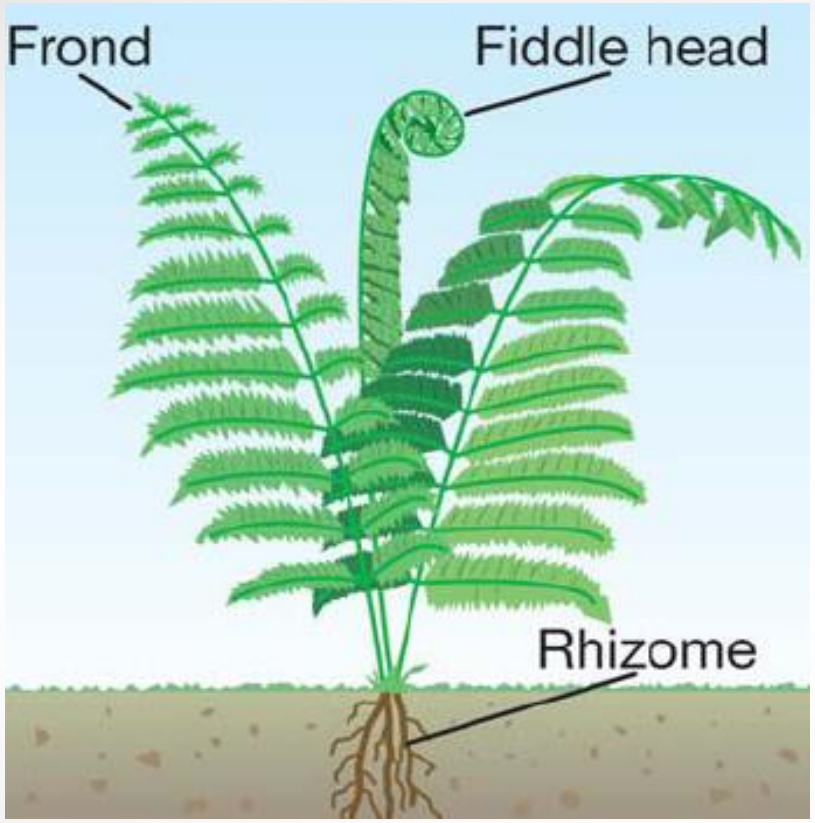
Club mosses



Horsetails

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Seedless Vascular plants

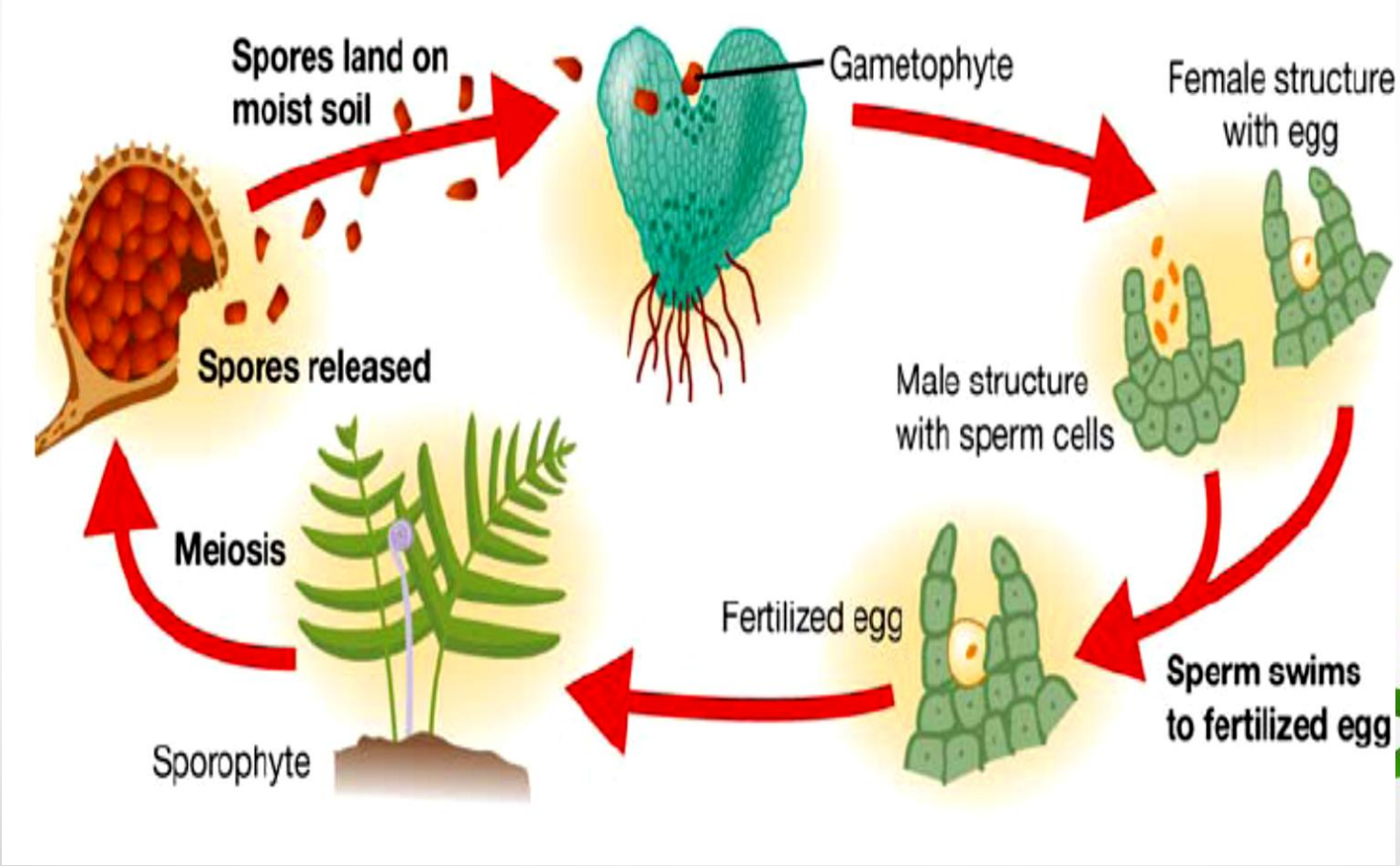


- The form of a fern is the sporophyte.
- Ferns have an underground stem called a **rhizome** from which the fronds unfurl.
- Young fronds are tightly coiled and are called **fiddleheads**.

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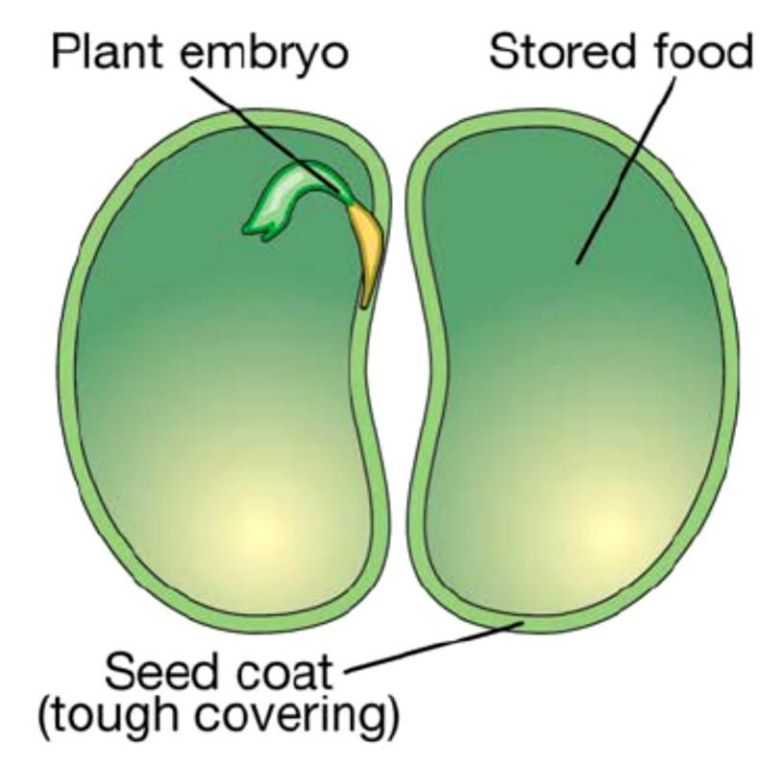
Seedless Vascular plants

- Ferns need water to transport sperm cells to egg cells.



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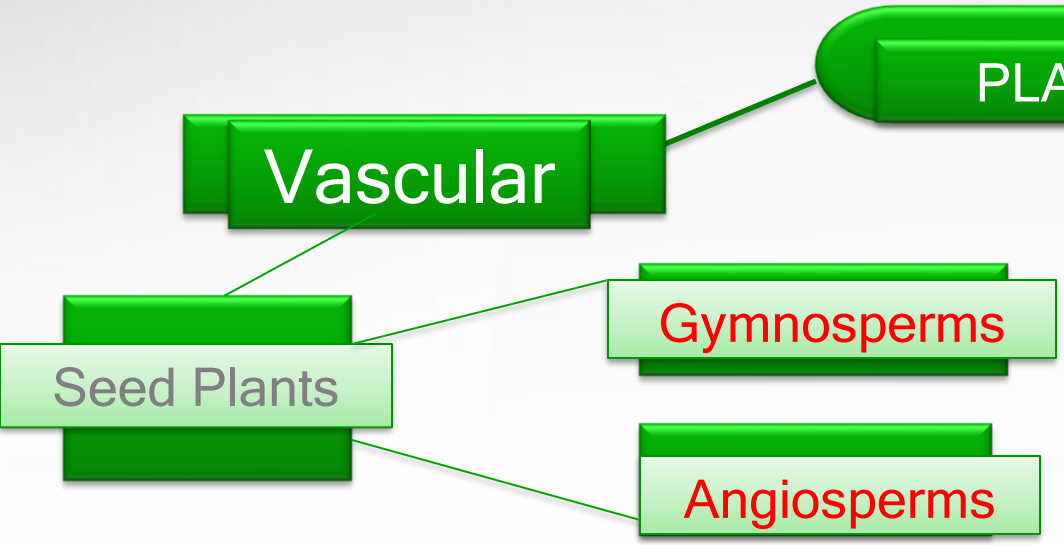
Vascular plants with Seed



- A seed is a structure that contains a plant embryo and a supply of food inside a protective covering.

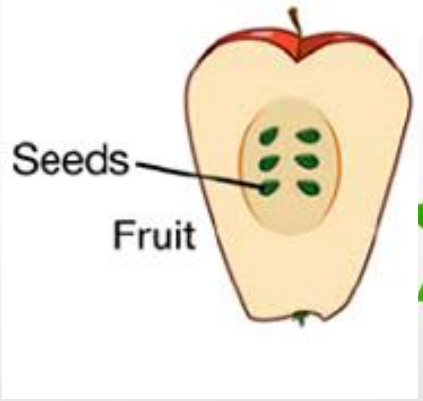
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Vascular plants with Seed



- Gymnosperms are a group of vascular plants whose seeds are housed in cones.
- Angiosperms, also known as flowering plants, produce seeds within a fruit.

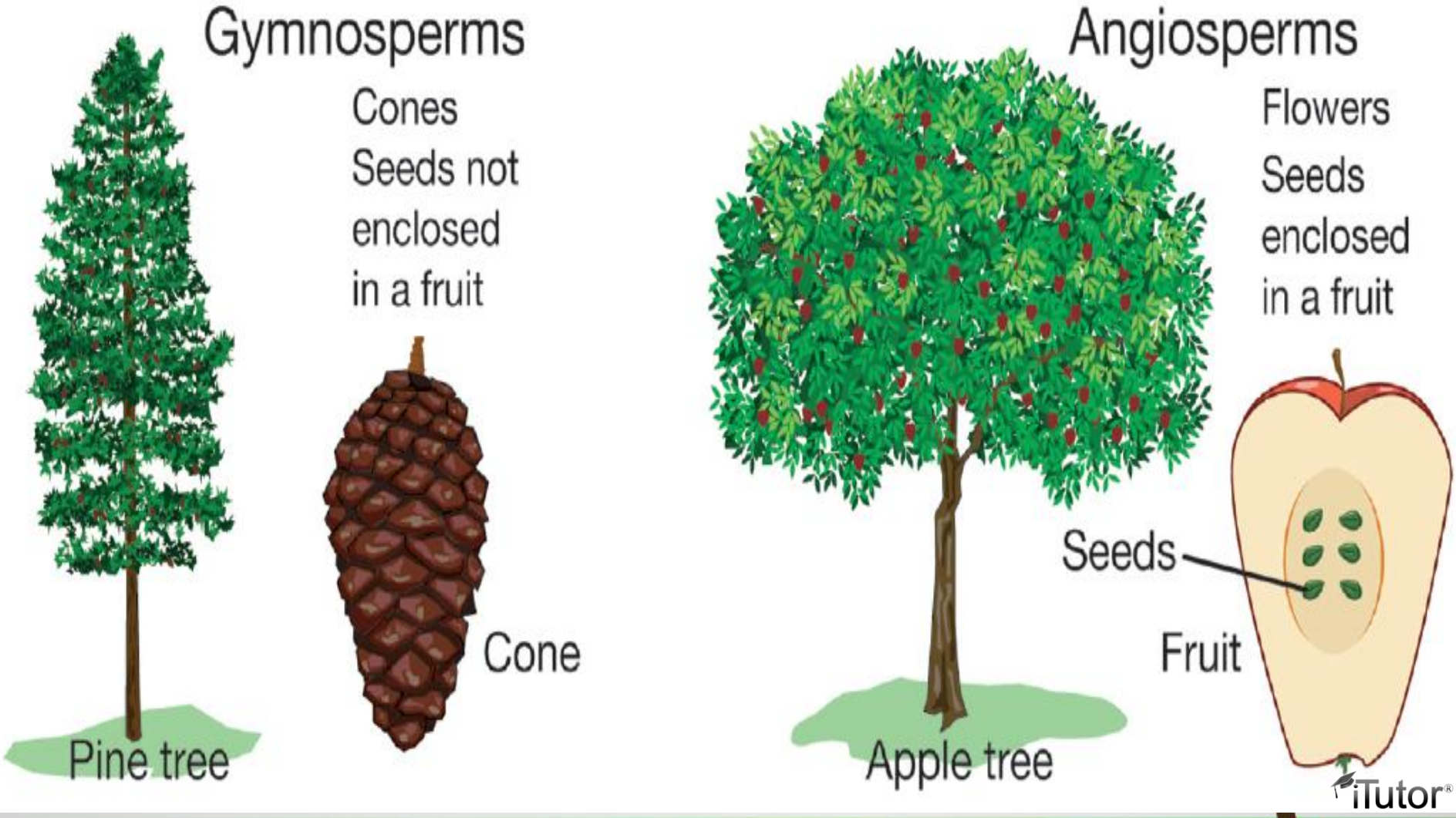






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Vascular plants with Seed



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Vascular plants with Seed

- Gymnosperms include conifers, cycads, and gingkoes.



- A gymnosperm called the bristlecone pine is the oldest living organism on Earth.

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Non - Vascular plants

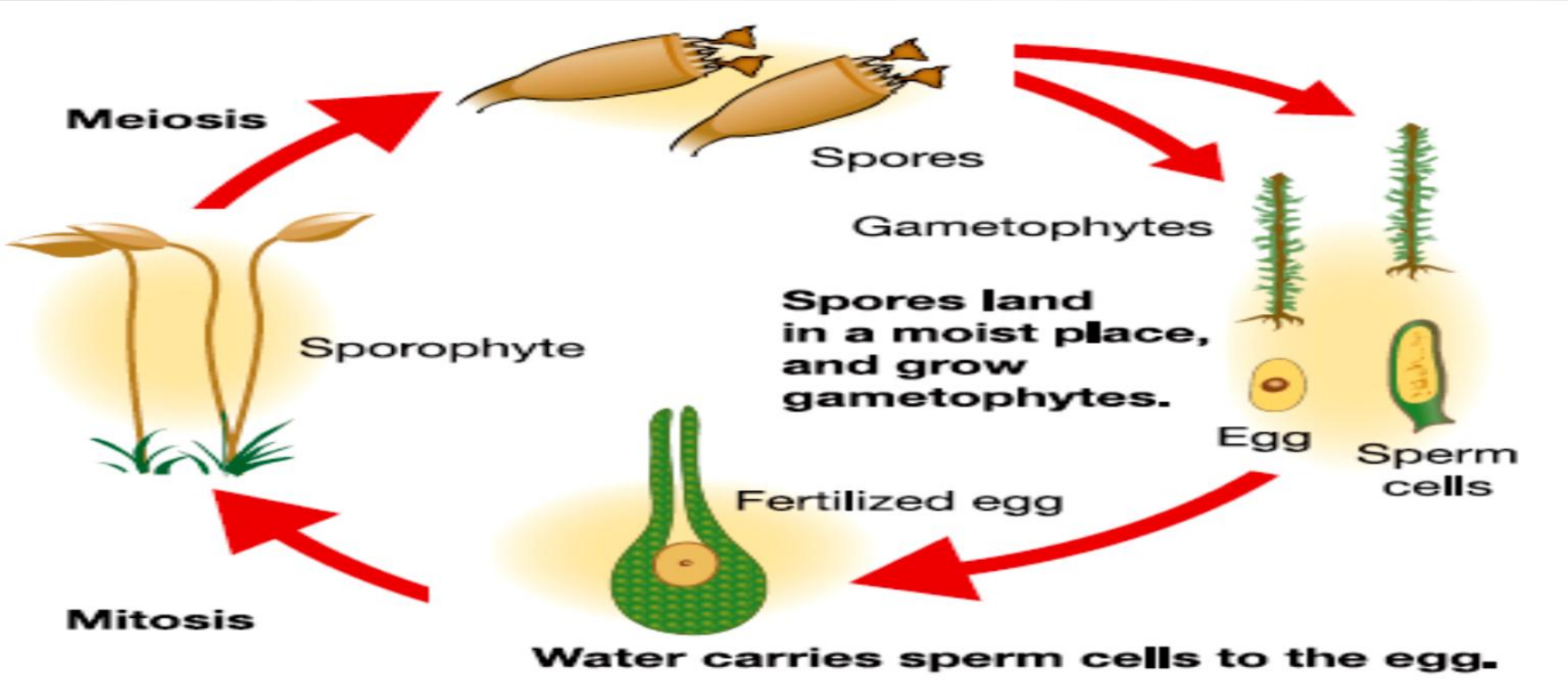
- The non-vascular plants include the mosses and liverworts.
- These are small, simple plants usually found in moist locations.





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Non - Vascular plants

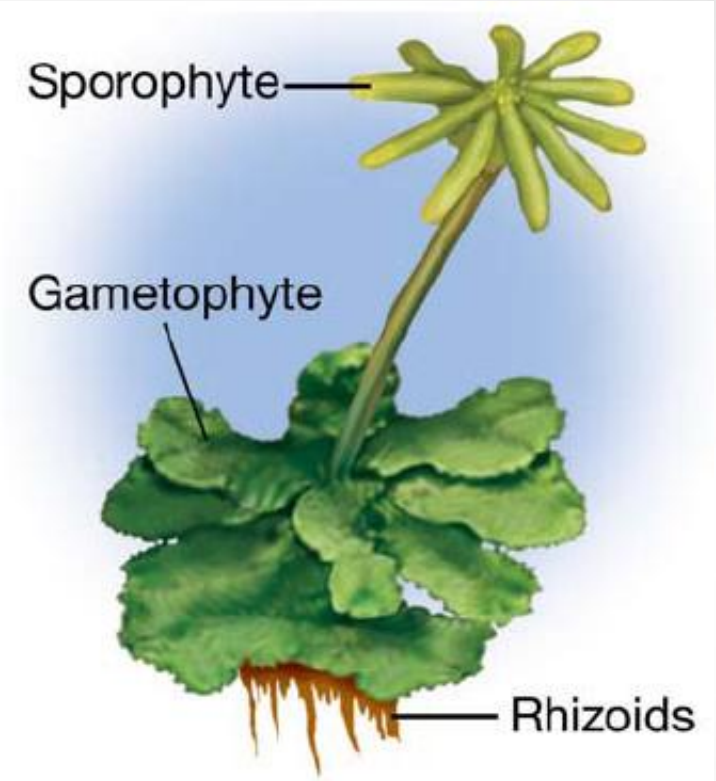


- The life cycle of nonvascular plants shows an alternation of generations.
- Mosses and liverworts need water to carry the sperm to the eggs for fertilization.

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Non - Vascular plants

- The sporophyte stage of a liverwort looks like a tiny palm tree.
- The body of the gametophyte stage is leafy and flattened.
- Rhizoids are root-like growths that extend from beneath the body and anchor the plant.



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Plant Kingdom – Plantae

Kingdom Plantae includes all the plants.

They are eukaryotic, multicellular and autotrophic organisms.

The plant cell contains a rigid cell wall.

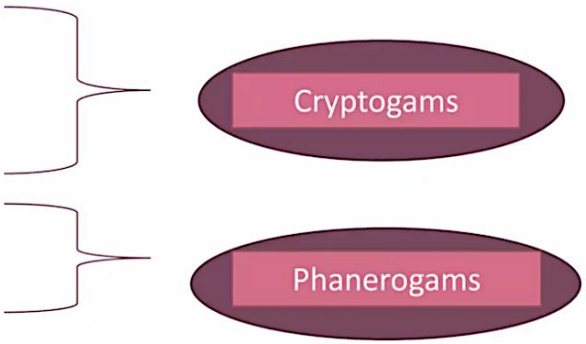
Plants have chloroplast and chlorophyll pigment, which is required for the photosynthesis

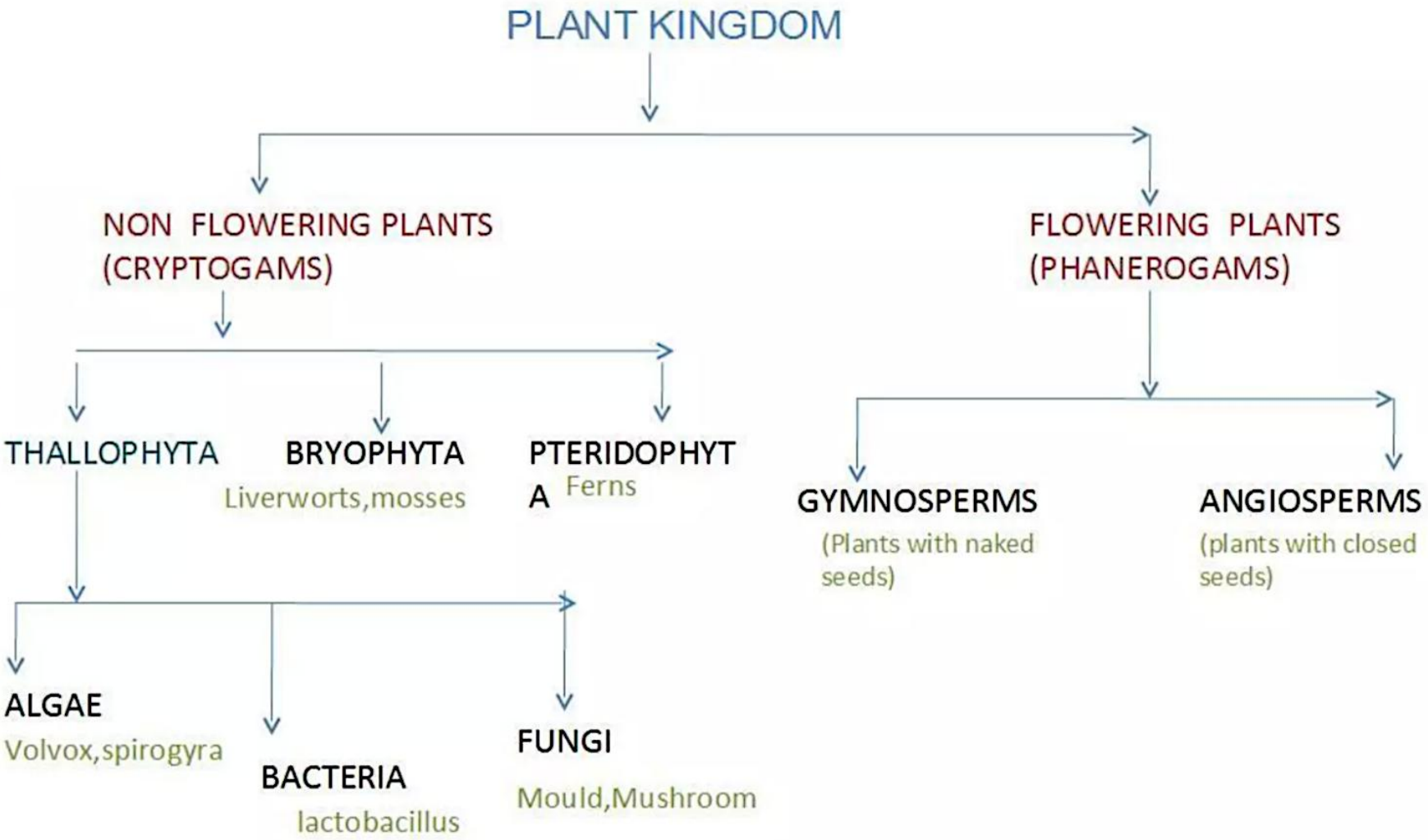


The plant kingdom is also classified into two groups:
Cryptogams – Non-flowering and non-seed bearing plants.

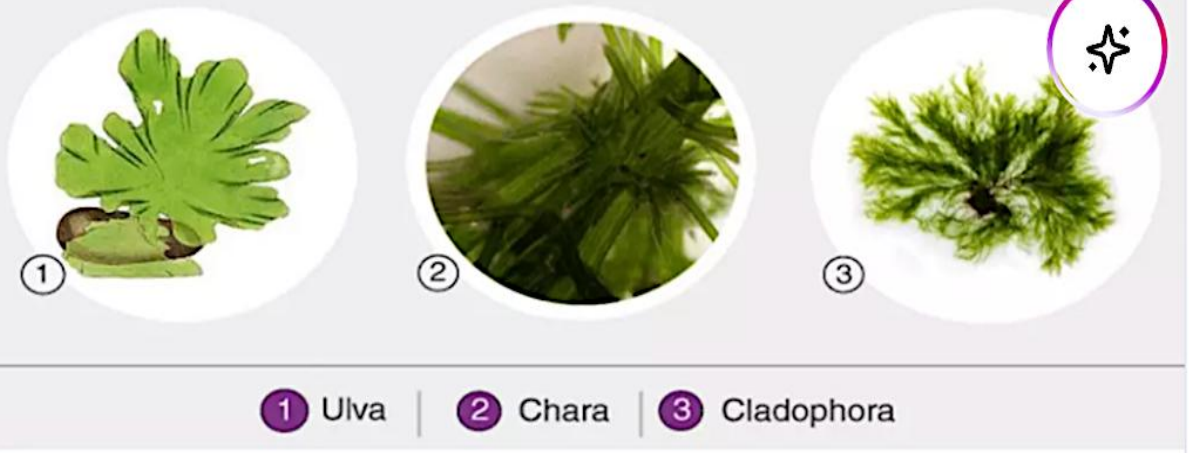
Phanerogams – Flowering and seed-bearing plants.

- The plant kingdom has been classified into five subgroups according to the above-mentioned criteria:
- Thallophyta
- Bryophyta
- Pteridophyta
- Gymnosperms
- Angiosperms




PLANT KINGDOM
NON FLOWERING PLANTS (CRYPTOGAMS)
FLOWERING PLANTS (PHANEROGAMS)
THALLOPHYTA
BRYOPHYTA
Liverworts,mosses
PTERIDOPHYTA Ferns
GYMNOSPERMS
(Plants with naked seeds)
ANGIOSPERMS
(plants with closed seeds)
ALGAE
Volvox,spirogyra
BACTERIA
lactobacillus
FUNGI
Mould,Mushroom

Thallophyta



- Thallophytes lack a well-differentiated body structure and the plant body is thallus like.
- Thallophytes: Primitive plants where the body is not differentiated into stem, roots and leaves

they may be filamentous, colonial, branched or unbranched.

Common examples are Volvox, Fucus, Spirogyra, Chara, Polysiphonia, Ulothrix, etc.

Characteristics of Thallophyta

- They are usually found in moist or wet places.
- This is due to the absence of “true roots” and vascular tissue that is needed to transport water and minerals. Hence they are found in moist or wet places.
- They are autotrophic in nature.
- Most members of this group manufacture their own food. But a few members like fungi are dependent on other sources of food.,
- Reserve food is generally starch.
- After photosynthesis, glucose is produced and consumed almost immediately, the remaining glucose is converted into complex compounds called starch.
- They have a cell wall composed of cellulose around their cells.
- Absence of vascular tissue.
- Unlike other plants, xylem and phloem are absent. etc
- Sex organs are simple, single-celled, there is no embryo formation after fertilization.





- Bryophytes do not have vascular tissues.
- The plant body has root-like, stem-like and leaf-like structures. Bryophytes are terrestrial plants but known as “amphibians of the plant kingdom” as they require water for sexual reproduction. They are present in moist and shady places.
- Bryophyta includes mosses, hornworts and liverworts. Some of the common examples are *Marchantia, Funaria, Sphagnum, Antheoceros*, etc

Characteristics of Bryophytes

- Plants in this category do not have roots but have crude stems and leaves.
- They have “rhizoids” instead of roots which helps the plant to anchor to surface.
- These roots or rhizoids do not absorb nutrients like other usual plant roots.
- Mosses release spores from their leaves which travels by water and make new mosses in new locations.
- Water is very essential for mosses to grow and spread. They can entirely dry out and survive. When in contact with water, they again revive and continue growing

Pteridophyta

- Pteridophytes have a well-differentiated plant body into root, stem and leaves. They have a vascular system for conduction of water and other substances. Some of the common examples are *Selaginella, Equisetum, Pteris*, etc.





characteristics of Pteridophytes

- Primary characteristics of Pteridophytes are as follows:
- They are seedless, vascular plants that show true alternation of generations. Furthermore, the sporophyte has true roots, stem and leaves.
- They reproduce by spores, which are developed in sporangia.
- They may be homosporous or heterosporous.



Gymnosperms

- Gymnosperms have a well-differentiated plant body and vascular tissues. They bear naked seeds, i.e. seeds are not enclosed within a fruit. Some of the common examples of gymnosperms are *Cycas, Pinus, Ephedra*, etc.



Angiosperms

- Angiosperms are seed-bearing vascular plants with well-differentiated plant body.
- The seeds of angiosperms are enclosed within the fruits.
- Angiosperms are widely distributed and vary greatly in size, e.g. Wolffia is small measuring about 0.1 cm and Eucalyptus trees are around 100 m tall.
- Angiosperms are further divided into monocotyledons and dicotyledons according to the number of cotyledons present in the seeds.
- Some of the common examples are mango, rose, tomato, onion, wheat, maize, etc.



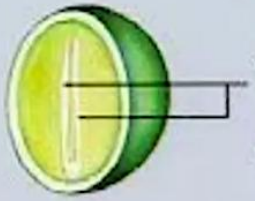
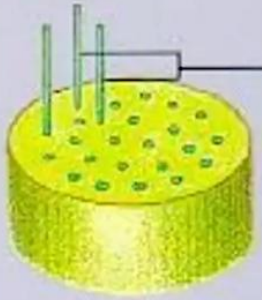
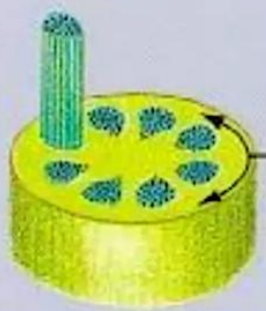


- Angiosperms are vascular plants with stems, roots, and leaves.
- The seeds of the angiosperm are found in a flower. These make up the majority of all plants on earth.
- The seeds develop inside the plant organs and form fruit. Hence, they are also known as flowering plants.
- Angiosperms are the most advanced and beneficial group of plants. They can grow in various habitats as trees, herbs, shrubs, and bushes.

Classification of Angiosperm

- **Monocotyledons**
- The seeds have a single cotyledon.
- The leaves are simples and the veins are parallel.
- This group contains adventitious roots.
- Each floral whorl has three members.
- It has closed vascular bundles and large in number.
- For eg., banana, sugarcane, lilies, etc.

- **Dicotyledons**
- The seeds of these plants have two cotyledons.
- They contain tap roots, instead of adventitious roots.
- The leaves depict a reticulate venation.
- The flowers are tetramerous or pentamerous and the vascular bundles are organized in rings.
- For eg., grapes, sunflower, tomatoes, etc

MONOCOTS	DICOTS
 one cotyledon	 two cotyledons
 floral parts in threes	 floral parts in fours or fives
 parallel leaf veins	 netlike leaf veins
 pollen grain has one pore or furrow	 pollen grain has three pores or furrows
 vascular bundles throughout stem's ground tissue	 stem's vascular bundles arranged in a ring